

# GeoZigzag Studio: A Reproducible Workflow for Agricultural Coverage and Semantic Georouting

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**Abstract**—Agricultural mission preparation spans two different spatial tasks: dense coverage of a field and sparse travel between geographically referenced targets. Existing workflows commonly prepare these tasks in separate tools, which complicates parameter tracking, route comparison, validation, and robot-oriented export. This paper presents GeoZigzag Studio, a reproducible workflow that combines boustrophedon coverage planning with semantic georouting in one web interface and Python evaluation package. Semantic georouting is defined as routing an ordered set of georeferenced targets while semantic land-cover labels influence path cost or validity. The implementation formalizes local projection, sweep-line coverage, direct interpolation, semantic-cost  $A^*$ , forbidden-zone checks, optional OSRM routing, heading generation, and ROS-style CSV/YAML export. A fixed-seed benchmark evaluates three fields and three mission scenarios. The fields span 141–3289 m<sup>2</sup> and generate 224–710 waypoints. Across routing scenarios, direct paths intersect 4–7 forbidden-zone segments; cost-aware paths reduce this count to zero with 4.6–7.3% additional distance. Cached OSRM routes expose maximum input-to-network snap distances of 35.2–90.8 m, demonstrating why network routes require explicit validation. The repository provides offline tests, versioned scenarios, cached service responses, generated figures and tables, and a single reproduction command.

**Index Terms**—agricultural robotics, coverage path planning, geospatial routing, reproducibility, waypoint preparation, ROS 2

## I. INTRODUCTION

Agricultural mobile robots operate at more than one spatial scale. Within a plot, sensing or treatment requires dense parallel passes. Between plots or resources, the mission is instead a sparse sequence of targets such as water points, crop areas, or access tracks. Coverage path planning (CPP) and geographic routing address these needs, but their inputs, controls, and output formats differ. Transferring a mission between tools can silently change coordinate order, sampling density, target order, or assumptions about traversability.

This separation also weakens experimental traceability. A screenshot alone does not identify the algorithm, route parameters, map state, or external service response that produced it. A research workflow should regenerate its routes, metrics, tables, and figures from versioned inputs. It should also distinguish a route that is geometrically valid from one that has been executed successfully by a robot.

GeoZigzag Studio addresses this preparation layer. It combines interactive field editing and mission ordering with a deterministic Python core. We use *semantic georouting* to mean:

given an ordered sequence of sparse WGS84 targets and a spatial semantic model, compute a connecting route for which labels such as track, shrub, or water modify traversal cost or admissibility. This differs from CPP, whose objective is dense traversal of a bounded area, and from local robot navigation, which closes the loop using localization and perception.

The contributions are:

- a unified, explicit workflow for dense field coverage and sparse semantic mission routing;
- formal algorithms for projection, coverage, cost-aware routing, forbidden-zone validation, and ROS-oriented orientation export;
- a multi-scenario benchmark comparing direct, semantic-cost, and cached OSRM routes using distance, waypoints, turns, time, intersections, snap distance, and success rate; and
- a reproducibility package with 20 offline tests, scenario files, service fixtures, generated artifacts, and `make reproduce`.

## II. RELATED WORK AND RESEARCH GAP

CPP surveys describe decomposition and back-and-forth patterns for complete area traversal [1], [2]. Agricultural CPP must additionally consider field shape, working width, headlands, and vehicle turning constraints [3], [4]. The Fields2Cover library provides reusable swath, route, and path components and a benchmarking basis [5]. Nilsson and Zhou proposed 54 standard fields and agronomic measures for systematic comparison [6]. GeoZigzag does not replace these richer vehicle-aware planners; it provides a lighter mission-preparation and reproduction layer.

General GIS tools such as QGIS can construct saved chains of vector and raster operations through the Processing Modeler [7]. OSMnx automates acquisition, projection, analysis, and routing over OpenStreetMap networks [8], while OSRM exposes network route and nearest services and reports the distance from an input coordinate to its snapped network location [9]. These tools are powerful, but they do not by themselves provide the combined agricultural coverage, semantic mission, and ROS-style export workflow considered here.

At execution time, Navigation2 provides ROS 2 planning and control and a waypoint-follower action [10], [11]. It expects a configured robot, frames, localization, costmaps,

TABLE I  
WORKFLOW-LEVEL CAPABILITY COMPARISON.

| Tool                   | CPP | Sem. | Web | Zones | ROS | Repro. |
|------------------------|-----|------|-----|-------|-----|--------|
| Fields2Cover           | Y   | –    | –   | P     | P   | Y      |
| QGIS                   | P   | P    | –   | Y     | –   | Y      |
| OSMnx/OSRM             | –   | Y    | P   | P     | –   | Y      |
| Nav2 waypoint follower | –   | –    | –   | Y     | Y   | P      |
| GeoZigzag Studio       | Y   | Y    | Y   | Y     | Y   | Y      |

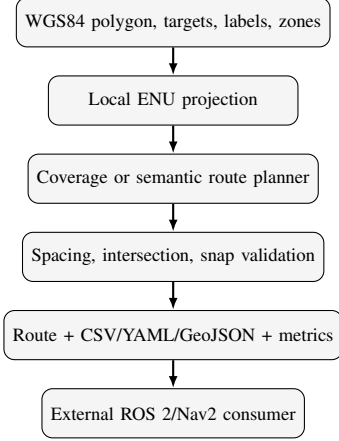


Fig. 1. GeoZigzag preparation pipeline. The external consumer is not part of the route-generation evaluation.

and controllers; it is a downstream consumer rather than a geographic authoring environment. Table I locates the gap. “P” means possible through configuration or plugins but not a first-class workflow; “ROS” denotes waypoint export or consumption, not validated execution.

### III. SYSTEM AND REPRODUCIBLE WORKFLOW

Figure 1 shows the software boundary. Inputs are (i) a WGS84 field polygon or labelled rectangle, (ii) row direction and start corner, (iii) positive row and waypoint spacings, (iv) an ordered set of WGS84 targets with semantic labels, (v) optional forbidden polygons, and (vi) a routing strategy. Outputs are the sampled route, yaw and planar quaternion at each point, CSV/YAML/GeoJSON files, validation status, metrics, and provenance.

The Leaflet interface makes generation explicit. Editing a field, spacing, target order, strategy, or forbidden zone marks the route *pending*, disables stale downloads, and does not silently regenerate. Generation ends in success, warning, or error state. OSRM warnings show the maximum snap distance; intersection failures and fallback details remain visible to the operator. Figure 2 shows both preparation modes.

#### A. Operator Workflow

In coverage mode, the operator draws or edits a rectangle or free polygon, selects a start corner and row bearing, and sets row and waypoint spacing. These two spacings remain independent: the former changes the agricultural swaths, while the latter changes the sampling density of the exported route. The field and previous route remain visible while parameters

are edited, but exports are marked stale until generation is requested again.

In mission mode, the operator creates an ordered sequence from GeoJSON targets or custom map points and selects direct, semantic-cost, or OSRM routing. Manual forbidden polygons encode areas that must be checked during preparation. An OSRM result is accompanied by its snap-distance warning; a cost-aware failure is reported rather than replaced silently by a direct segment.

After generation, the map provides the spatial sanity check, the summary shows route-scale metrics, and the status block distinguishes success, warning, and error. CSV is intended for inspection and analysis, YAML for ROS-oriented consumers, and GeoJSON for independent GIS verification. The experiment runner uses the same planners and schemas without the browser, binding every reported result to a versioned scenario configuration.

## IV. METHODS

### A. Projection, Heading, and Orientation

For field-scale geometry, all input coordinates remain WGS84 while computation uses a local equirectangular ENU approximation. For latitude–longitude  $(\phi, \lambda)$  and origin  $(\phi_0, \lambda_0)$  in radians,

$$x = R \cos \phi_0 (\lambda - \lambda_0), \quad y = R (\phi - \phi_0), \quad (1)$$

with  $R = 6,378,137$  m,  $x$  east, and  $y$  north. The origin is the first polygon point for coverage and the mean input coordinate for routing. This approximation is used only for the sub-kilometre scenarios evaluated here.

For consecutive local points  $p_i, p_{i+1}$ , ROS ENU yaw is  $\psi_i = \text{atan2}(\Delta y, \Delta x)$  normalized to  $[-\pi, \pi]$ . The final point reuses the previous yaw. Exported planar orientation is  $(q_x, q_y, q_z, q_w) = (0, 0, \sin(\psi/2), \cos(\psi/2))$ .

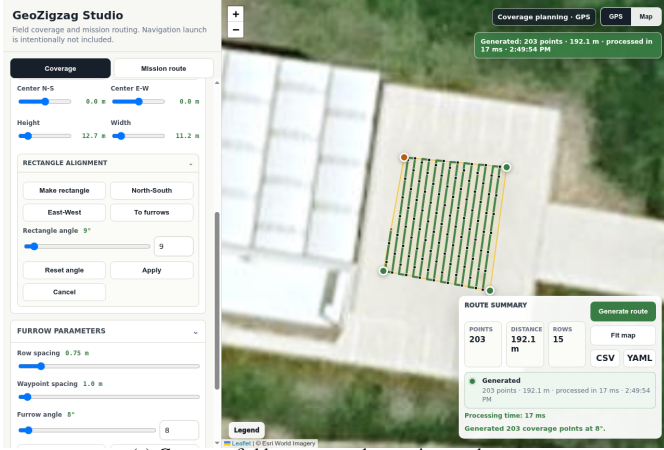
### B. Coverage Planner

The sweep direction is selected explicitly or inferred from the longest polygon edge. Perpendicular sweep lines are separated by row spacing  $s_r$ . Each line is intersected with every polygon edge; sorted intersections are paired into valid in-field intervals. Samples along an interval are no farther apart than waypoint spacing  $s_w$ . Traversal direction alternates by sweep.

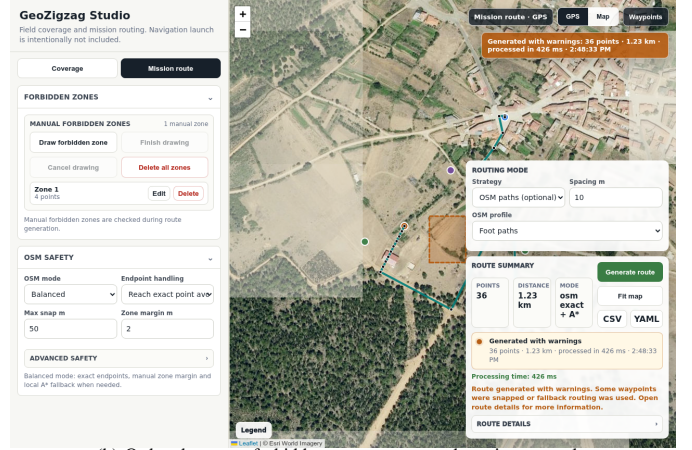
The implementation reports polygon area, row count, mean clipped row length, waypoint count, route distance, and discrete turns. A “turn” is a heading change of at least  $30^\circ$ ; it is a polyline metric, not a claim of kinematic feasibility.

### C. Semantic and Network Routing

Direct routing samples straight segments between ordered targets. Cost-aware routing rasterizes all inputs on an 8-connected local grid. The evaluated resolution is 4–5 m. A semantic point influences a radius of two cells (four for water), using costs 1, 1.5, 10, 30, 80, 1000 for road, track, grass, shrub,



(a) Coverage field, row controls, preview, and export.



(b) Ordered targets, forbidden zone, status, and routing controls.

Fig. 2. The two GeoZigzag Studio route-preparation modes. Both use explicit generation, a shared route summary, visible validation state, and the same waypoint export schema.

#### Algorithm 1: Polygon zigzag coverage

- 1: validate polygon,  $s_r > 0$ , and  $s_w > 0$
- 2: project vertices to local ENU and choose sweep axes
- 3: **for**  $d = \min d_{\perp}$  to  $\max d_{\perp}$  in steps  $s_r$
- 4:   intersect line  $d$  with all polygon edges
- 5:   sort intersections and pair inside intervals
- 6:   sample each interval at maximum spacing  $s_w$
- 7:   append samples forward/reverse on alternating sweeps
- 8: attach yaw/quaternion; return route and coverage metrics

#### Algorithm 2: Semantic/cost-aware mission route

- 1: validate ordered targets, resolution, labels, and zones
- 2: project inputs; build padded local cost grid
- 3: assign semantic costs; mark forbidden cells  $\infty$
- 4: **for each** consecutive target pair  $(g_i, g_{i+1})$
- 5:   run 8-connected A\* using cost  $J$
- 6:   **if** no path: return explicit failure
- 7:   restore exact target endpoints and concatenate
- 8: test every route segment against every zone edge
- 9: attach orientation; export route, validation, and metrics

scrub, and water. Forbidden polygon cells receive infinite cost. For a path  $P = (v_0, \dots, v_n)$ , A\* minimizes

$$J(P) = \sum_{k=1}^n c(v_k) \ell(v_{k-1}, v_k), \quad (2)$$

where  $\ell$  is one or  $\sqrt{2}$  grid steps. The admissible heuristic is  $c_{\min}$  times Euclidean cell distance. Failure to connect a target pair is reported; there is no silent direct-route fallback.

For optional network routing, the OSRM route service returns a network geometry and snapped waypoint records. GeoZigzag records  $d_{\text{snap}} = \max_i d(g_i, \hat{g}_i)$  and rejects results above a configured threshold. The offline benchmark uses a versioned response cache, not a live service, so results do not change with network availability or OSM updates.

Forbidden-zone validation counts each route segment that lies inside, touches, or crosses at least one polygon. It combines endpoint-in-polygon tests with orientation-based segment intersection, including collinear contact. A route is valid with respect to declared zones only when this count is zero.

#### D. Validation State and Failure Semantics

Let  $H(P, \mathcal{Z})$  be the number of segments in route  $P$  that intersect any polygon in the declared zone set  $\mathcal{Z}$ , and let  $S(P)$  be the maximum OSRM snap distance when network routing is used. Given configured snap limit  $S_{\max}$ , the route-level acceptance predicate is

$$\mathcal{V}(P) = [|P| \geq 2] \wedge [H(P, \mathcal{Z}) = 0] \wedge [S(P) \leq S_{\max}]. \quad (3)$$

The snap term is omitted for local routes. This predicate describes only the declared planning constraints; it is not a physical safety certificate.

The software distinguishes four states. *Pending* means an input changed after the last generation, so existing downloads are disabled. *Success* means the selected planner returned a route satisfying its configured checks. *Warning* preserves an inspectable result while exposing conditions such as nonzero but permitted snap distance. *Error* returns no current export when spacing is invalid, geometry is degenerate, targets are missing, a route is impossible, or a required service response is unavailable. The status and detailed reason are stored separately so the main map remains readable without hiding the failure cause.

Export occurs only from a non-pending generated state. CSV and YAML contain the same WGS84, yaw, and quaternion values; GeoJSON adds a route line and indexed point features. This one-to-one schema permits programmatic equality checks across formats and prevents a browser preview from diverging silently from the files used in evaluation.

## V. EXPERIMENTAL EVALUATION

### A. Research Questions

The benchmark is organized around four questions. *RQ1*: do row spacing and waypoint spacing produce distinguishable and predictable effects on coverage geometry and serialization density? *RQ2*: what distance and path-complexity penalty is paid when declared forbidden zones are enforced? *RQ3*: how different are network-routed missions from exact-target

TABLE II  
SEMANTIC ROUTING SCENARIO CONFIGURATION. TARGET INITIALS  
DENOTE WATER (W), SHRUB (A), GRASS (P), AND SCRUB (M).

| Scenario        | Order | Grid (m) | Sample (m) | Zones |
|-----------------|-------|----------|------------|-------|
| north-loop      | W-A-W | 5        | 10         | 1     |
| west-east       | P-M-P | 5        | 10         | 1     |
| hazard-crossing | P-M-W | 4        | 8          | 1     |

TABLE III  
COVERAGE RESULTS GENERATED FROM THE CONFIGURED FIELDS.

| Scenario          | Area (m <sup>2</sup> ) | Rows | Points | Dist. (m) |
|-------------------|------------------------|------|--------|-----------|
| compact_rectangle | 141.2                  | 16   | 224    | 205.8     |
| medium_rectangle  | 932.0                  | 27   | 347    | 682.0     |
| irregular_field   | 3288.9                 | 36   | 710    | 1753.7    |

direct routes, particularly in input-to-network snap distance? *RQ4*: can every reported route, metric, figure, and table be regenerated offline from versioned inputs?

These questions evaluate workflow behavior rather than ranking GeoZigzag against full vehicle-aware CPP or navigation systems. *RQ1* is addressed by the coverage scenarios and sensitivity sweep; *RQ2* by paired direct and cost-aware routes; *RQ3* by cached OSRM responses; and *RQ4* by repeated execution, artifact generation, and automated tests.

### B. Protocol

The evaluation configuration fixes seed 20260703 and executes each primary planner five times. Timing is the median wall-clock duration on an Intel Core i7-13700KF with Python 3.10; it excludes file export and plotting. Three coverage scenarios represent a compact rectangle, a medium quadrilateral, and an irregular five-vertex field. Their areas are 141.2, 932.0, and 3288.9 m<sup>2</sup>. Three semantic scenarios use different target orders and manual forbidden-zone layouts. Every routing scenario is evaluated with direct, semantic-cost A\*, and cached OSRM routing. The OSRM driving responses were captured on 3 July 2026 and retain their reported snap distances.

Metrics are waypoint count, distance, coverage rows, heading-change turns, median computation time, forbidden-zone intersections, maximum OSRM snap distance, and success rate. Coverage sensitivity crosses four row spacings (0.5, 0.75, 1.0, 1.5) m with three waypoint spacings (0.5, 1.0, 2.0) m.

### C. Results

All 60 primary executions (12 planner-scenario combinations, five repeats) completed, for 100% software-level generation success. Coverage produced 224, 347, and 710 waypoints over 16, 27, and 36 rows, with route distances of 205.8, 682.0, and 1753.7 m. Median coverage computation remained below 1 ms on the test machine.

Table IV shows the routing comparison. Direct routing is the shortest exact-target baseline but intersects declared zones. Cost-aware routing eliminates every intersection, increasing distance by 4.7%, 4.6%, and 7.3% for north-loop, west-east, and hazard-crossing missions.

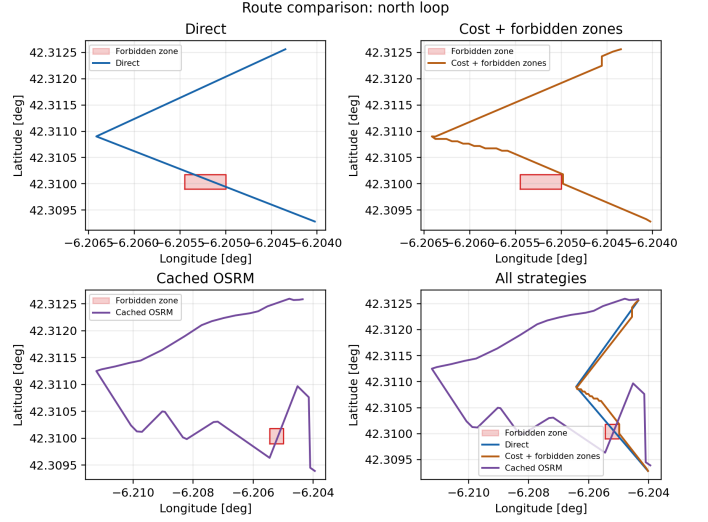


Fig. 3. Direct, forbidden-zone-aware semantic-cost, cached OSRM, and overlay views for the north-loop mission.

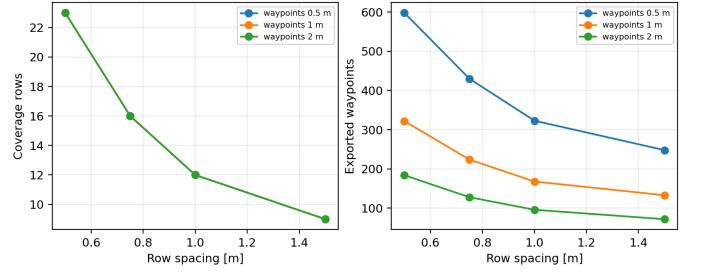


Fig. 4. Sensitivity to row and waypoint spacing on the compact field.

and hazard-crossing missions. Its larger turn count exposes grid discretization and shows why these routes still require smoothing or a local planner before execution.

Cached OSRM follows the mapped driving network rather than the straight mission geometry. It is substantially longer in two scenarios and still intersects zones in two. More importantly, maximum snap distance is 87.7, 90.8, and 35.2 m. Thus a successful HTTP response is not sufficient evidence that a network route reaches an agricultural target. Figure 3 visualizes this difference.

Sensitivity results in Fig. 4 separate geometric coverage from sampling density. Increasing row spacing from 0.5 to 1.5 m reduces rows from 23 to 9 and distance from 291.1 to 120.6 m. At fixed 0.75 m row spacing, changing waypoint spacing from 0.5 to 2.0 m changes exported points from 430 to 128 while preserving route distance (205.8 m to numerical precision).

### D. Robustness and Negative Cases

Table V summarizes failure-oriented checks. They execute offline and assert observable behavior, not only that an exception occurred. In particular, an impossible cost-grid mission is rejected rather than silently converted into a direct route, and

TABLE IV  
ROUTING RESULTS GENERATED BY `make reproduce`. “HITS” COUNTS ROUTE SEGMENTS INTERSECTING DECLARED FORBIDDEN ZONES.

| Scenario        | Strategy    | Points | Dist. (m) | Turns | Hits | Time (ms) |
|-----------------|-------------|--------|-----------|-------|------|-----------|
| north_loop      | direct      | 54     | 518.2     | 1     | 4    | 0.05      |
| north_loop      | cost_aware  | 83     | 542.7     | 18    | 0    | 59.38     |
| north_loop      | osrm_cached | 34     | 1552.0    | 11    | 1    | 0.02      |
| west_east       | direct      | 54     | 522.8     | 1     | 5    | 0.05      |
| west_east       | cost_aware  | 98     | 546.8     | 12    | 0    | 22.93     |
| west_east       | osrm_cached | 30     | 1125.2    | 6     | 0    | 0.02      |
| hazard_crossing | direct      | 42     | 314.7     | 1     | 7    | 0.04      |
| hazard_crossing | cost_aware  | 66     | 337.7     | 16    | 0    | 83.30     |
| hazard_crossing | osrm_cached | 11     | 293.6     | 4     | 3    | 0.01      |

TABLE V  
OFFLINE ROBUSTNESS CHECKS AND REQUIRED OUTCOMES.

| Input or event               | Required outcome                |
|------------------------------|---------------------------------|
| $s_r \leq 0$ or $s_w \leq 0$ | Reject before planning          |
| Degenerate polygon           | No coverage export              |
| Target enclosed by a zone    | Explicit no-route error         |
| OSRM timeout/failure         | External-service error          |
| Edit after generation        | Pending state; exports disabled |
| Excessive OSRM snap          | Reject above configured limit   |

TABLE VI  
VERSIONED INPUTS AND GENERATED REPRODUCIBILITY ARTIFACTS.

| Artifact          | Reproducibility role                  |
|-------------------|---------------------------------------|
| evaluation.yaml   | Scenarios, parameters, seed, repeats  |
| osrm_fixture.json | Cached geometry, source, date, snaps  |
| waypoints.*       | CSV/YAML route and orientation schema |
| route.geojson     | Independent GIS route inspection      |
| results.csv       | Per-scenario metrics and success rate |
| summary.json      | Config hash, provenance, full results |
| figures/*.png     | Code-generated paper comparisons      |

an OSRM transport failure is reported through the external-routing error contract.

## VI. DISCUSSION

The experiments expose three workflow-level trade-offs. First, row spacing and waypoint spacing are not interchangeable controls. Row spacing changes the physical sweep geometry, route length, and number of turns; waypoint spacing primarily changes serialization density. Reporting only waypoint count would therefore confound two different design decisions.

Second, a shorter geographic route is not necessarily an admissible route. Direct interpolation provides a useful lower bound on distance but crosses the declared zones in all three routing scenarios. The cost-aware planner removes those intersections at a measured distance penalty, while its additional corners reveal the need for curvature-aware post-processing. Validation and path quality must consequently be reported together.

Third, network routing introduces a different uncertainty. The OSRM service successfully returned all cached routes, yet two routes were much longer than the direct alternatives and snap distances reached 90.8 m. Keeping exact input targets, snapped geometry, service provenance, and validation results separate prevents a visually plausible map route from being mistaken for an executable robot path.

## VII. REPRODUCIBILITY AND SOFTWARE VALIDATION

The command `make reproduce` runs 20 offline tests, the configured evaluation, figure/table generation, and LaTeX compilation. Each successful route is exported as CSV, YAML, and GeoJSON. The summary records the random seed, configuration SHA-256 digest, fixture version, per-run success, and all reported metrics. Tests cover rectangle and polygon coverage, invalid spacing, yaw/quaternion conversion,

direct and cost-aware routing, impossible routes, forbidden intersections, export schemas, mocked OSRM success/failure, generated files, and web-state contracts. Online map tiles and live OSRM are optional and excluded from offline tests.

The Python modules separate geometry, coverage, routing, metrics, export, and evaluation. The previous public API remains as a compatibility facade. This keeps the web/CLI workflow stable while allowing each scientific component to be exercised independently.

Visual verification is part of the protocol but not substituted for numeric validation. Each generated GeoJSON route can be overlaid with its targets, field boundary, and forbidden zones. This catches coordinate-order or origin errors that may still produce syntactically valid CSV/YAML files. Conversely, the intersection and snap metrics prevent a visually plausible screenshot from being treated as sufficient evidence.

### A. Downstream ROS 2 and Simulation Boundary

The export is a preparation contract, not a complete launch interface. A Nav2 consumer must convert WGS84 coordinates into its configured map frame, preserve the exported orientation convention, and validate localization and costmap assumptions. Likewise, camera-based crop-row control consumes images and publishes velocity commands rather than geographic waypoints.

Repository inspection identified a feasible staged validation: generate a Gazebo world with the selected geographic origin, spawn the agricultural robot, verify camera, odometry, command, and TF interfaces, then alternate global row-entry goals with local crop-row control. This experiment requires explicit command arbitration and row-end transitions; it is kept separate from the software benchmark so that preparation metrics are not presented as navigation performance.

### VIII. LIMITATIONS AND THREATS TO VALIDITY

The semantic cost layer is a controlled proxy built from labelled point neighborhoods, not a calibrated traversability map. The grid paths contain discrete corners and do not enforce turning radius, headland width, vehicle footprint, soil response, or dynamics. Manual forbidden zones are only as complete as operator input. The local projection is appropriate for the short routes studied here, not regional missions.

Cached OSRM data improve repeatability but cannot establish current network quality. Live results may change, fail, or snap differently as OSM and server profiles evolve. Timing values are machine-dependent and the scenario set is too small for claims of general superiority. The benchmark measures route preparation, not localization, control, obstacle avoidance, simulated completion, or field performance. ROS 2 waypoint execution and camera-based crop-row following require separate frame, topic, controller, and safety validation. Future work should compare against Fields2Cover on standard field sets and measure executed cross-track error, missed area, interventions, completion time, and target arrival error in simulation and on a robot.

### IX. CONCLUSION

GeoZigzag Studio turns two commonly separated planning tasks into a measurable and reproducible agricultural mission-preparation workflow. Dense boustrophedon coverage and sparse semantic georouting share projection, validation, orientation, export, and provenance. The evaluation demonstrates three concrete findings: row and waypoint spacing affect different properties; declared forbidden zones can be avoided at a measurable distance and complexity cost; and OSRM snap

distance must be reported rather than hidden. The resulting artifact generates its routes, metrics, figures, and manuscript tables from versioned inputs while keeping robot execution claims outside the evidence provided.

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